Snap: Chlorophyll Concentration Calculator using RAW Images of Leaves

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***Abstract*—Determination of nitrogen content, a major component of chlorophyll in plant leaves, is a demand. In this paper we propose *Snap*, a handheld optical sensor device that uses RAW images of leaves for determining their chlorophyll content. By skipping the pre-processing pipeline at image signal processor, RAW images help to generate a relatively linear relationship between the incident illumination and the calculated pixel value. Pixel values are used to calculate spectral parameters, which are known for their correlation with the chlorophyll concentration. Illumination on the camera lens is controlled by specific geometrical constraints, which are derived using the diffusion properties of the illumination sources. The apparatus banishes influence of external illumination. The approach, by integration with a mobile app, caters to spatial variation of nutrients in soil. A pilot implementation involving 50 cotton plants was conducted for comparing the spectral-parameter value calculated by *Snap* and SPAD 502 plus Chlorophyll meter. A correlation factor of 0.908 was observed, thereby proving significant similarity but at a tremendously lower cost.**

Keywords*—Digital Cameras; RAW images; Nutrient determination; Spectral Parameter; ground-based remote sensing.

I. INTRODUCTION

Fertilizers account for half of the world's food produce [1]. However, an overly enthusiastic use of fertilizers has rather dire consequences and plants are unable to tolerate acidification of soil. This results in "haying off" in crops such as wheat [2]. Leafy crops tend to have higher nitrogen concentrations in an over-fertilized field, affecting consumers' health with gastric and bladder cancer [3]. While under-fertilization does not allow for maximum productivity, over-fertilization has implications affecting the entire society [4]. Farmers' heuristic approach of deploying a standard amount of fertilizer per unit area has skewed the soil nutrient pattern, which should have a Nitrogen: Phosphorus ratio of 4:1, ideally, but has rocketed to 24:1 in India and 300:1 in China [5].

Soil testing is an inescapable need of the hour. The currently adopted methodologies for soil analysis include high-precision laboratory solutions based on chemical analysis or handy, low-precision electrochemical/optical sensors. Laboratory solutions (such as, Duma's method) use a destructive methodology that tends to be time consuming and ignores the spatial variation of soil nutrient profile [6]. Meanwhile, electrochemical relate nitrate conductivity to soil nitrogen concentration in a cost-effective manner. However, the cumbersome deployment methodology and ignorance of nitrite and ammonium ions' concentration has limited their implementation. Optical handheld meters, such as SPAD and FieldScout [7][8], are relatively new solutions and use narrow-bandwidth, relevant range (680-710 nm and 750-850 nm) photodiodes to calculate remote sensing based spectral parameters—such as, NDVI (normalized difference vegetation index). Red and NIR (near infrared) illumination sources are used to illuminate a leaf and the transmittance or reflectance from the leaf is used to decide nitrogen content as a function of chlorophyll concentration [9]. These devices are highly expensive, but easy to use.

The aforementioned limitations of different methodologies necessitate the need of a new solution. It has been demonstrated that digital cameras are capable of performing accurate analysis (within constricted environments) of many leaf properties such as nutrient/water deficiencies ([10], [11]), senescence/growth stage [12], insect damage [13], and growth parameters like LAI (leaf area index) [14]. Affordability and use of optical characterization strategies serve as incentives for us to explore digital cameras as a viable alternative for automating soil analysis. The existing methodologies involving digital cameras tend to depend on heuristics, than formal strategies. Thus, we have devised *Snap*, a handheld optical sensor based upon RAW images and GNDVI (green normalized difference vegetation index), a spectral parameter appropriate for ground-based remote sensing analysis, to calculate leaf-nitrogen content. Its name is a repercussion of using images as the basis of analysis.

II. LIMITATIONS TO USING DIGITAL CAMERAS FOR LEAF-NUTRIENT CALCULATION

A commercially available digital camera and an image processing system are ubiquitous to most embedded systems. Thus, a methodology based upon digital cameras to calculate leaf-nutrient content is easily deployable.

Past work has shown that significant correlations (R=0.85-0.90) between leaf chlorophyll content and RGB-derived parameters of color-images are attainable [15], [16]. However, variation in sunlight with respect to time induces inherent variability in the correlation model. To solve the same, researchers have used heuristics, such as capturing images at fixed times of the day to ensure that illumination and cloudiness of the scenes are similar [17]. Geometrical variations in image-capture methodology, i.e., distance of the camera from the leaf and diffusion characteristics of the illumination sources, lead to unreliability and are dealt with using unreliable heuristics [12]. We cater to these problems by devising a self-sufficient apparatus with internal, specifically located illumination sources.

Digital cameras pre-process images using the ISP (image signal processor) pipeline. Thus, the RGB values achieved are not in direct accord with the incident intensity on the camera lens. Vesali [18] demonstrated the statistical unreliability

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induced as a repercussion of image pre-processing. To overcome the same, Sakamoto [14] modeled a 6th order empirical equation and achieved better results as compared to the direct RGB analysis. We provide a relatively robust solution by using RAW images, which are unprocessed digital data outputted by the image sensor.

III. PROPOSED METHODOLOGY

We cater to the limitations found in the existing methodologies by using an apparatus that ensures geometrical consistency and banishes external illumination from influencing the analysis. Further, RAW images have been used to remove the influence of ISP-pipeline and remote sensing based spectral parameters have been modified to account for quantum efficiency of the image sensor.

A. *Snap Apparatus (shown in Fig. 1):*

Fig. 1 shows a three dimensional view of the proposed apparatus, named *Snap*. A camera holding apparatus ensures geometrical consistency while sensing reflected illumination. The dimensions are based upon the diffusion characteristics of illumination sources used (see Fig. 2), thereby ensuring that the amount of energy incident at each point is the same for various wavelengths. An anti-reflectant is painted on the walls to ensure captured energy is reflected only from the leaf.

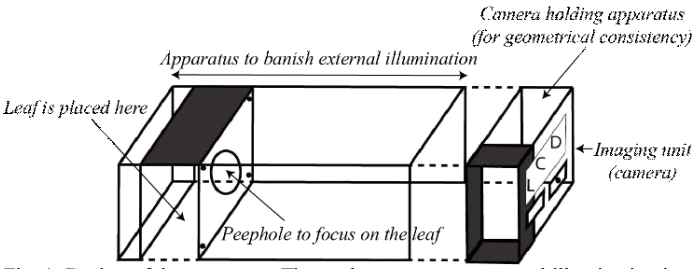


Fig. 1. Design of the apparatus. The enclosure ensures external illumination is unable to affect the derived correlation. The dimensions are calculated as per Fig. 2.

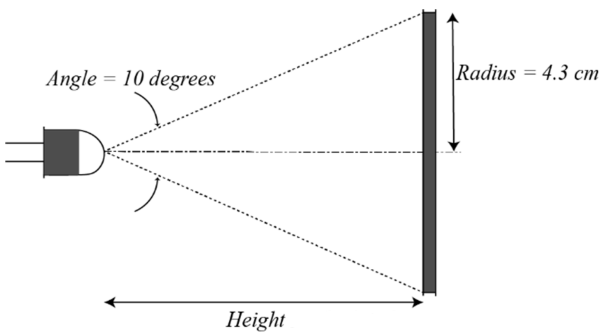


Fig. 2. Representative diagram of diffuse illumination of an LED, used for calculating height of the prototype's apparatus. The diffusion characteristics are used to ensure that different wavelengths have similar amount of energy incident at each point. A circular region of radius 4.3 cm was illuminated by LEDs with half angle of 10 degrees, resulting in a corresponding apparatus height (mentioned in figure) of 24.38 cm.

B. *Spectral Parameter used:*

GNDVI proclaims better correlation with chlorophyll concentration. Therefore, the same should be used in ground based remote sensing analysis. We use the mathematical formulation from Gitelson et al. [19], as shown in (1).

$$GNDVI = \frac{r(NIR) - r(green)}{r(NIR) + r(green) - 2.r(blue)} \quad (1)$$

In (1), $r(x)$ denotes reflectance, i.e., percentage of energy reflected, in the illumination region corresponding to x. As per the relevant range for GNDVI, we use narrowband LEDs with of wavelengths 570 nm, 460 nm, and 850 nm.

C. *Parameter modification to account for the spectral response of the sensor:*

As shown in [20], spectral parameters require a modification in order to account for the quantum efficiency of an image sensor, at a given wavelength. Expanding on the same, we have devised "effective quantum efficiency" (defined by (2)) to account for the spectral composition of an illumination source and relate it with the spectral response of the image sensor.

$$EFQ = \int_{\lambda 1}^{\lambda 2} \frac{q(\lambda).\lambda.RI(\lambda)d\lambda}{\int_{\lambda 1}^{\lambda 2} RI(\lambda)d\lambda} \quad (2)$$

In (2), q(λ) is the quantum efficiency of the image sensor at wavelength "λ" and RI(λ)dλ is the percentage of emitted radiant intensity in the wavelength span dλ. Using the effective quantum efficiency and pixel values of a RAW image, one can approximate the incident energy on the image sensor by dividing the RGB pixel value with the corresponding effective quantum efficiency. This results in a modified GNDVI, as defined in (3).

$$modGNDVI = \frac{\frac{P(NIR)}{EFQ} - \frac{P(Green)}{EFQ(Green)}}{\frac{P(NIR)}{EFQ(NIR)} + \frac{P(Green)}{EFQ(Green)} - \frac{2P(Blue)}{EFQ}} \quad (3)$$

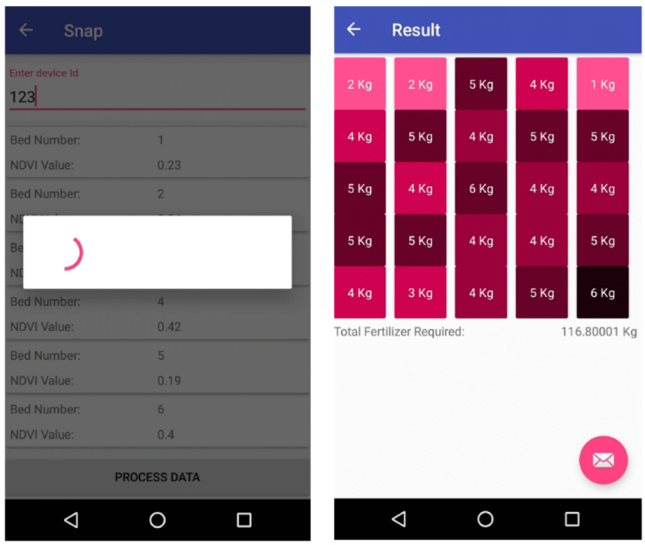


Fig. 3. GUI of the Android application and depiction of a generated heat map. The heat map details spatial variation of nutrients in soil.

D. *Use of RAW images for linear analysis:*

A RAW image is unprocessed data readout from the image sensor. The data is yet to go through the non-linear digital

preprocessing pipeline at the ISP. Therefore, digital number values at each pixel are linear with respect to incident energy on the image sensor. This direct accord between RAW images' pixel values and incident intensity helps improve machine-learning models' ability for better generalization. Using EXIF tool, we reverse any unnecessary quantization and extract the bayer pattern of the image sensor. Using this information, the image signal is segregated into different color channels and modified-GNDVI is calculated.

E. Spatial variation of soil profile:

Our methodology, being a handheld and easy to carry device, allows us to account for spatial variation of soil profile. We have designed an android application (see Fig. 3) which takes calculated spectral parameters and GPS location as inputs for creating a heat-map of required and optimum fertilizer inputs.

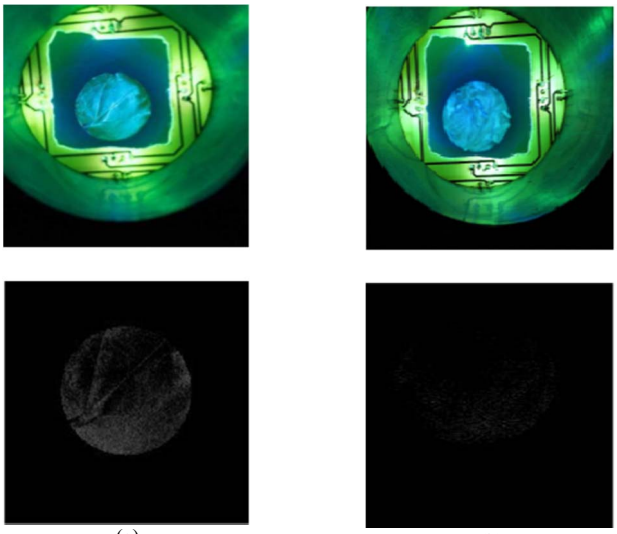
Fig. 4. Final working prototype.

IV. EXPERIMENTAL SETUP

For our test analysis, we use a Nikon B70 to model the proposed methodology (see Fig. 4). Comparing the images of a healthy and dead leaf captured using *Snap*, we can observe significant differences between the 2-D representation maps of GNDVI values calculated at each pixel (as shown in Fig. 5). The density of white pixels in the processed images relate to the amount of chlorophyll present in that region.

V. RESULTS

We conducted an initial round of analysis of our proposed methodology on leaves of various health ranges. Using 50 samples of Cotton leaves, a linear regression model was generated between the parameter GNDVI, as calculated by *Snap*, and SPAD 502 plus Chlorophyll meter's calculated NDVI values. Since no existing device uses GNDVI as its basis, GNDVI and NDVI had to be correlated. Thereafter, 24 samples were obtained for validation and a correlation of R=0.908 was observed. The NDVI values for the 24 samples, as calculated using SPAD 502 plus Chlorophyll meter and *Snap*'s linear regression model, have been plotted in Fig. 6. Our results show that at similar accuracy, *Snap* is a viable alternative to existing chlorophyll meters.



(a) (b)

Fig. 5. 2-D representation maps of calculated spectral parameter (GNDVI) at each pixel using *Snap*. Fig. 5(a) corresponds to a moderately healthy leaf (Fig. 5(a)) and Fig. 5(b) corresponds to a dead leaf.

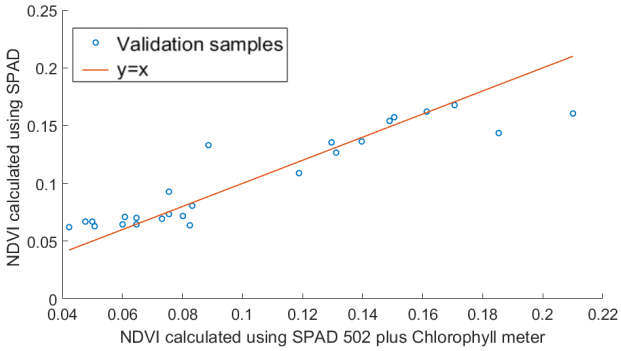


Fig. 6. Analysis of the Linear Regression model that correlates GNDVI values from *Snap* and SPAD 502 plus Chlorophyll meter. 24 test samples have been plotted.

VI. CONCLUSION

In this paper, we described *Snap*, a handheld optical sensor for calculating leaf chlorophyll concentration. The proportionality of nitrogen and chlorophyll concentration help bridge several limitations to currently existing soil-nutrient analysis methods. A geometrical apparatus that banishes external illumination brings about geometrical consistency to *Snap*. Further, use of RAW images and modification of spectral parameters based upon the image sensor's quantum efficiency make the entire methodology robust. Connecting the device with an android application, we generate a heat-map to account for spatial variation of nutrients in soil. A pilot implementation correlating *Snap* and SPAD 502 plus Chlorophyll meter's measured values showed significant correlation, thereby proving *Snap* to be a viable and cheaper alternative. Future work will focus on generating correlation algorithms for the calculated value of the spectral parameter and nutrients, such as Phosphorus, Magnesium and Potassium. They relate well with chlorophyll concentration of differently located leaves, and can be easily analyzed using *Snap* [21].

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